

Q1 - 26 July - Shift 1

The disintegration rate of a certain radioactive sample at any instant is 4250 disintegrations per minute. 10 minutes later, the rate becomes 2250 disintegrations per minute. The approximate decay constant is :

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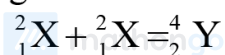
(Take $\log_{10} 1.88 = 0.274$)

(A) 0.02 min^{-1} (B) 2.7 min^{-1}

(C) 0.063 min^{-1} (D) 6.3 min^{-1}

Q2 - 26 July - Shift 2

Two lighter nuclei combine to form a comparatively heavier nucleus by the relation given below:



The binding energies per nucleon ${}^2_1\text{X}$ and ${}^4_2\text{Y}$ are 1.1 MeV and 7.6 MeV respectively. The energy released in this process is _____ MeV.

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Q3 - 27 July - Shift 1

What is the half-life period of a radioactive material if its activity drops to $1/16^{\text{th}}$ of its initial value of 30 years ?

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(A) 9.5 years (B) 8.5 years

(C) 7.5 years (D) 10.5 years

Q4 - 27 July - Shift 2

Questions

MathonGo

The activity of a radioactive material is 6.4×10^{-4} curie. Its half life is 5 days. The activity will become 5×10^{-6} curie after :

- (A) 7 days (B) 15 days
(C) 25 days (D) 35 days

*Space for your notes:***Q5 - 28 July - Shift 1**

The half life period of a radioactive substance is 60 days. The time taken for $\frac{7}{8}$ th of its original mass to disintegrate will be :

- (A) 120 days (B) 130 days
(C) 180 days (D) 20 days

*Space for your notes:***Q6 - 28 July - Shift 1**

A freshly prepared radioactive source of half life 2 hours 30 minutes emits radiation which is 64 times the permissible safe level. The minimum time, after which it would be possible to work safely with source, will be _____ hours.

*Space for your notes:***Q7 - 28 July - Shift 2**

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Questions

MathonGo

A radioactive sample decays $\frac{7}{4}$ times its original quantity in 15 minutes. The half-life of the sample is

- (A) 5 min (B) 7.5 min
(C) 15 min (D) 30 min

*Space for your notes:***Q8 - 29 July - Shift 2**

Two radioactive materials A and B have decay constants 25λ and 16λ respectively. If initially they have the same number of nuclei, then the ratio of the number of nuclei of B to that of A will be "e" after a time $\frac{1}{a\lambda}$. The value of a is _____.

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Q1 (C)At $t=0$ disintegration rate = 4250 dpmAt $t=10$ disintegration rate = 2250 dpm

$$A = A_0 e^{-\lambda t}$$

$$2250 = 4250 e^{-\lambda(10)}$$

$$\Rightarrow \lambda(10) = \ln \left(\frac{4250}{2250} \right)$$

$$\Rightarrow \lambda = 0.063 \text{ min}^{-1}$$

Q2 (26)

Energy released in the given process = Binding energy of product – Binding energy of reactants

$$= 7.6 \times 4 - (1.1 \times 2) \times 2$$

$$= 30.4 - 4.4$$

$$= 26 \text{ MeV}$$

Q3 (C)

$$A = A_0 e^{-\lambda t}$$

$$\Rightarrow -\lambda t = \ln\left(\frac{A}{A_0}\right)$$

$$\Rightarrow -\frac{\ln 2}{t_{1/2}} \times 30 = \ln\left(\frac{1}{16}\right)$$

$$\Rightarrow -\frac{\ln 2}{t_{1/2}} \times 30 = -4 \ln 2$$

$$\Rightarrow t_{1/2} = \frac{30}{4} = 7.5 \text{ yrs}$$

Q4 (D)

$$A_0 = 6.4 \times 10^{-4} \text{ Curie}$$

$$T_{1/2} = 5 \text{ days} = \frac{\ln 2}{\lambda}$$

$$A = A_0 e^{-\lambda t}$$

$$5 \times 10^{-6} = 6.4 \times 10^{-4} e^{-\lambda t}$$

$$\frac{5}{6.4} \times 10^{-2} = e^{-\lambda t}$$

$$7.8 \times 10^{-3} = e^{-\lambda t}$$

$$\log(7.8 \times 10^{-3}) = -\lambda t \ln e$$

$$\ln(7.8 \times 10^{-3}) = -\frac{\lambda \ln 2}{5} \cdot t$$

$$\therefore \frac{5 \times 4.853}{0.693} = t = 35 \text{ days}$$

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Q5 (C)

$7/8$ disintegrates means $1/8$ remains

$$\text{Or } \left(\frac{1}{2}\right)^3$$

$\therefore$ 3 half lives

$= 180$ days

Q6 (15)

$$A = A_0 \times 2^{-t/T}$$

$$\frac{A_0}{64} = A_0 \times 2^{-t/T}$$

$$\therefore t = 6T = 6 \times 2.5 = \boxed{15} \text{ hours}$$

Q7 (A)

$$\frac{I}{C} \times \text{area} = \text{force}$$

$$\frac{I}{C} \times 36 \times 10^{-4} = 7.2 \times 10^{-9}$$

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Q8 (9)

$$N = N_0 e^{-\lambda t}$$

$$\frac{N_B}{N_A} = \frac{e^{-\lambda_2 t}}{e^{-\lambda_1 t}} = e^{-\lambda_2 t} \cdot e^{\lambda_1 t}$$

$$e^1 = e^{(\lambda_1 - \lambda_2)t}$$

$$(\lambda_1 - \lambda_2)t = 1$$

$$t = \frac{1}{\lambda_1 - \lambda_2} = \frac{1}{25\lambda - 16\lambda} = \frac{1}{9\lambda}$$

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